When does a bulb glow?

Read this before the C3 initial micro-test.

STUDENT READING

1. The parts of a simple circuit

A simple circuit can contain an electric cell, connecting wires, a bulb and a switch. The cell provides electrical energy. The wires connect the components. The bulb has two electrical contact points, and the wires must connect to those contact points rather than to the glass.

2. A complete conducting path

A circuit is complete when there is an unbroken conducting path from one terminal of the cell, through the connected components, and back to the other terminal. In a simple working circuit, this path includes both electrical contacts of the bulb. When the path is complete and the components are working, current can flow and the bulb can glow.

Having every component is not enough. The components must be connected correctly. A wire that is loose, a connection made to the wrong part of a bulb, or a break anywhere in the path can stop current from flowing through the circuit.

3. Open and closed circuits

A closed circuit has a continuous conducting path. An open circuit has a gap or break in that path. A switch controls the path: when it is closed (ON), its conducting parts touch, so the switch no longer interrupts the path; when it is open (OFF), a gap remains at the switch. The whole circuit works only if the rest of the path is connected too. A switch can be placed at different points in the circuit; its job is still to make or break the path.

4. Explain the observation, not just the result

If a bulb does not glow, do not immediately conclude that the cell is dead or the bulb is faulty. The switch may be open, a wire may be loose, a connection may touch the wrong place, or a component may not work. Check the path and the components systematically. One observation usually does not identify the cause by itself.

Remember these ideas

- A complete circuit is an unbroken conducting loop between the two terminals of a cell.
- Both electrical contacts of the bulb must be connected into the path.
- An open switch creates a gap; a closed switch removes that gap, but cannot fix breaks elsewhere in the circuit.
- If the bulb stays off, use evidence and careful checks before deciding why.

Safety reminder: this lesson is about low-voltage cells and classroom components. Do not test ideas using household electrical outlets.